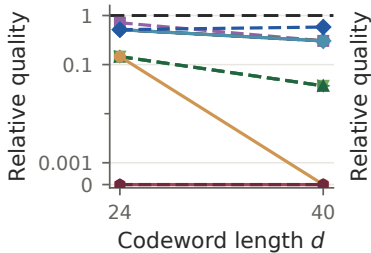
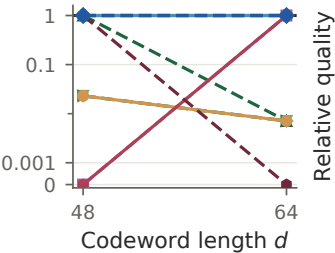


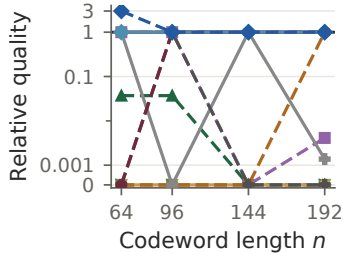
Shannon  $C_7$   
2 sizes



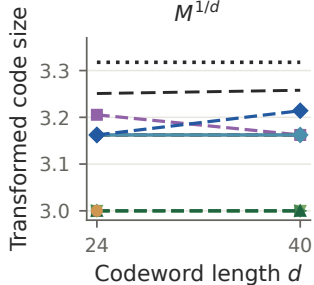
Shannon  $C_9$   
2 sizes



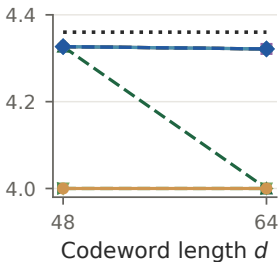
Trifference  
4 sizes ( $m = 1$ )



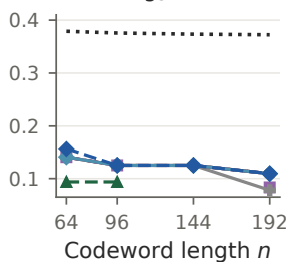
$M^{1/d}$



$M^{1/d}$



$\log_3 M/n$



— Qwen3.5-4B high  
— Qwen3.5-9B high  
— Qwen3.5-27B high  
— Qwen3.8-27B high  
— GPT-5.6 Luna medium

— GPT-5.6 Luna high  
— DeepSeek V4.1 Flash low  
— DeepSeek V4.1 Flash high  
— Claude Fable 5.1 medium  
— Claude Fable 5.1 high

— GPT-6 Astra medium  
— GPT-6 Astra high  
— Reference  
— Proven bound